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Secure Hybrid Shortest Path Algorithm in Mobile Ad hoc Networks for Optimize Link State Routing

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ABSTRACT
MANETs are the significant technologies among various air communication technologies where all the nodes are mobile and can be connected dynamically using wireless link in a random manner. All nodes work router and maintain routes to other nodes in the network. In this recent research work, a new system called, Secure Hybrid Shortest Path (SHSP) Routing compare with optimized link state routing (OLSR) proactive routing protocol is designed for MANETs to reduce end-to-end delay, secure hybrid shortest path routing algorithm besides avoiding delay and packet loss, it increases the node speed also. Network Simulator (NS2) is used to simulate the proposed method and implemented in the test system. The new routing protocol named OLSR-SHSP using OLSR is proposed to address the problem. Shortest path system based transmission is highly secure with the lowest delay and packet dropping. All the three scenarios OLSR-SHSP provides better performance compared to the existing system routing protocol by decreasing end-to-end delay, lowering routing overhead and reducing packet drop compared to the OLSR routing protocol. To enhance the merits of this research work, there is a plan to investigate the following issues in the future. However, the same concept can be applied in satellite to reduce end-to-end delay in the route and reduce packet loss, Possibilities of adopting secure quality oriented techniques to further improve the network performance of quality.

Keywords: *MANET, Routing, Delay, OLSR, SHSP*

INTRODUCTION
A directing convention determines how the switches speak with each other, dispersing data that empowers them to choose courses between any two hubs on a PC organization. Directing calculations decide the particular decision of course. Every switch just has an earlier information on networks connected to it straightforwardly. A steering convention divides this data first between prompt neighbors, and afterward all through the organization. Directing conventions are made for switches. These conventions have been intended to permit the trading of directing tables among switches. The switch finds out about remote organizations from neighbor switches or from an executive and constructs a steering table. In the event that the organization is straightforwardly associated, the switch definitely knows how to get in to the organization. Subsequently, to save the power utilization, course handing-off load, battery duration, and decrease in the recurrence of sending control messages, streamlining of the size of control headers, an effective course reconfiguration ought to be thought about while fostering a steering convention.[1-7]

ROUTING PROTOCOL
A static directing convention determines how switches speak with one another, dispersing data that empowers them to choose courses between any two dynamic hubs on

a PC organization. Steering calculations decide the particular decision of course. Every switch just has an earlier information on networks connected to it straightforwardly. Steering conventions divide this data between other prompt neighbors, and afterward entire organization. The two main types of routing are static routing and dynamic routing. The whole switch finds out about remote organizations from neighbor switches or from an executive and fabricates an update directing table. If the network is directly connected, then the router already knows how to get to the network. The router learns how to get to the remote network with either static or dynamic routing. If static routing is used, then the administrator has to change all routers in the network and therefore no routing protocol is used. By and large, there are two unique stages in steering: they are course revelation and sending information bundles. In course disclosure, the course to an objective will be found by communicating the question. Then, at that point, when the un-fragile course has been laid out, information sending will be started and sent through the courses that not entirely set in stone. The power utilization, course transferring load, battery duration, and decrease in the recurrence and transmission capacity of sending control messages, improvement of size of headers and proficient dynamic course reconfiguration ought to be thought about while fostering a steering convention. In this paper utilizing OLSR-SHSP proactive steering convention.[8-10]

PROBLEM DEFINITIONS

The unique idea of MANETs requires the steering conventions to revive the steering tables habitually and experience the ill effects of transmission dispute time postponement and clog with bundle dropping that are the consequences of the telecom idea of radio transmission since a hub in MANETs can't straightforwardly speak with the hubs outside its correspondence range, a parcel might need to be directed through transitional hubs to arrive at the objective. Subsequently it becomes vital for screen the imperatives in middle hubs. Subsequently, an effective directing methodology might produce course delay, parcel dropping and course disappointments. The least difficult plan directing in MANETs is the one in tracking down a course without malignant hubs in the most brief way. This paper intends to give strong course to protect transmission with the briefest way. So a new directing calculation named OLSR-SHSP utilizing OLSR with secure hy-brid most limited way is proposed. This OLSR-SHSP gives better execution contrasted with the current framework and furthermore lessens directing postponement and parcel dropping with no misbe-haviour at middle hubs. The unique idea of MANETs requires the steering conventions to revive the steering tables habitually and experience the ill effects of transmission dispute time postponement and clog with bundle dropping that are the consequences of the telecom idea of radio transmission since a hub in MANETs can't straightforwardly speak with the hubs outside its correspondence range, a parcel might need to be directed through transitional hubs to arrive at the objective. Subsequently it becomes vital for screen the imperatives in middle hubs. Subsequently, an effective directing methodology might produce course delay, parcel dropping and course disappointments. The least difficult plan directing in MANETs is the one in tracking down a course without malignant hubs in the briefest way.

This paper intends to give strong course to protect transmission with the briefest way. So a new directing calculation named OLSR-SHSP utilizing OLSR with secure hybrid most limited way is proposed. This OLSR-SHSP gives better execution contrasted with the current framework and furthermore lessens directing postponement and parcel dropping with no misbehaviour at middle hubs.[11-15]

PROPOSED CONCEPT

Secure Hybrid Shortest Path (SHSP) Routing Algorithm

S-Source Node, D-Destination Node, Q-Queue, T-Traffic SeLi is the security level of node and SeLp is the security level in the RREQ packet {The Destination node sends RREP back}

Step 1: Source node broadcasts a RREQ to all its neighbours Repeat for neighbour nodes.

Step 2: If there is a route to the destination node then authenticate the RREQ.

Step 3: Calculate its security level using Secure Routing. If SeLi > SeLp

Step 4: Update the security level in the RREQ packet overwrite the SeL in RREQ packet with efficient model Else

Step 5: Broadcast the RREQ to its neighbour nodes.

Step 6: If network = traffic then Q queue check to D Else

Step 7: HSPR $\Longrightarrow$ D {Hybrid shortest path routing model} Else if Occurred traffic T

Step 8: If network ≠ T Route HSPR to D Else

Step 9: Dropped Packets {data loss model} Find the other route of the network.

Step 10: Using the HSPR algorithm model S data send to D on network.

Step 11: Best route path selection method process over.

SIMULATION CONFIGURATIONS

The simulator NS 2.34 is used to test the optimized link state routing protocol, and as shown in Table 1, the following are the simulation parameters used for the analysis of routing protocol with hybrid shortest path algorithm.

Table 1: *Simulation parameters.*

Parameter	**Values**
Examined protocol	OLSR-SHSP
Application traffic	CBR
Transmission range	800m
Packet size	512 bytes
Maximum speed	25m/s
Simulation time	700s
Number of nodes	125
Area	800m x 800m

RESULTS AND DISCUSSION

In this research work, compare performance of OLSR and OLSR-SHSP test and analysis with five different parameters.

Table 2: *Packet delivery ratio.*

RP / NN	**25**	**50**	**75**	**100**	**125**
OLSR	**0.86**	**0.82**	**0.73**	**0.68**	**0.66**
OLSR-SHSP	**0.92**	**0.86**	**0.77**	**0.72**	**0.70**

From Table 2, it is clear that secure proposed scheme OLSR-SHSP surpasses OLSR performance by above 74% when there are 25 and 125 nodes in the network.

Table 3: *Routing overhead.*

RP / NN	**25**	**50**	**75**	**100**	**125**
OLSR	**0.15**	**0.28**	**0.35**	**0.44**	**0.58**
OLSR-SHSP	**0.12**	**0.25**	**0.32**	**0.40**	**0.55**

Simulation results of routing overhead shown in Table 3. It is clear that OLSR-SHSP has the lowest overhead of about 25 to 125 number of nodes then existing system.

Table 4: *Throughput.*

RP / NN	**25**	**50**	**75**	**100**	**125**
OLSR	**0.15**	**0.27**	**0.40**	**0.53**	**0.57**
OLSR-SHSP	**0.25**	**0.37**	**0.50**	**0.63**	**0.68**

Table 4 proves that the proposed OLSR-SHSP provides better performance of the throughput when there are 25 to 125 of nodes compared to OLSR routing protocol.

Table 5: *Remaining energy.*

RP / NN	**25**	**50**	**75**	**100**	**125**
OLSR	**0.92**	**0.84**	**0.76**	**0.72**	**0.64**
OLSR-SHSP	**0.95**	**0.89**	**0.80**	**0.76**	**0.70**

Table 5 clearly show that the proposed OLSR-SHSP increases the remaining energy with increasing number of nodes from 25 to 125 compared to OLSR.

Table 6: *End-to-end delay*

RP / NN	**25**	**50**	**75**	**100**	**125**
OLSR	**0.17**	**0.23**	**0.29**	**0.38**	**0.47**
OLSR-SHSP	**0.12**	**0.18**	**0.24**	**0.33**	**0.42**

According to table 6, it is clear that proposed scheme OLSR-SHSP surpassed the performance of OLSR in minimising end-to-end delay by 7% when there are 25 to 125 nodes in the network. As the proposed algorithm finds different short routes frequently, it is possible to minimize the delay.

From all the above tables, it is clear that the comparison of the OLSR-SHSP illustrate that the proposed algorithm outperforms the OLSR by providing lowest end-to-end delay, packet drop and routing overhead with increase in the number of nodes.

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